

SMART INDIA HACKATHON 2026



TITLE PAGE

PROJECT: Baghewala Digital Twin | CSS + SRP operations

Problem Statement ID: [ENTER EXACT PORTAL PS ID]

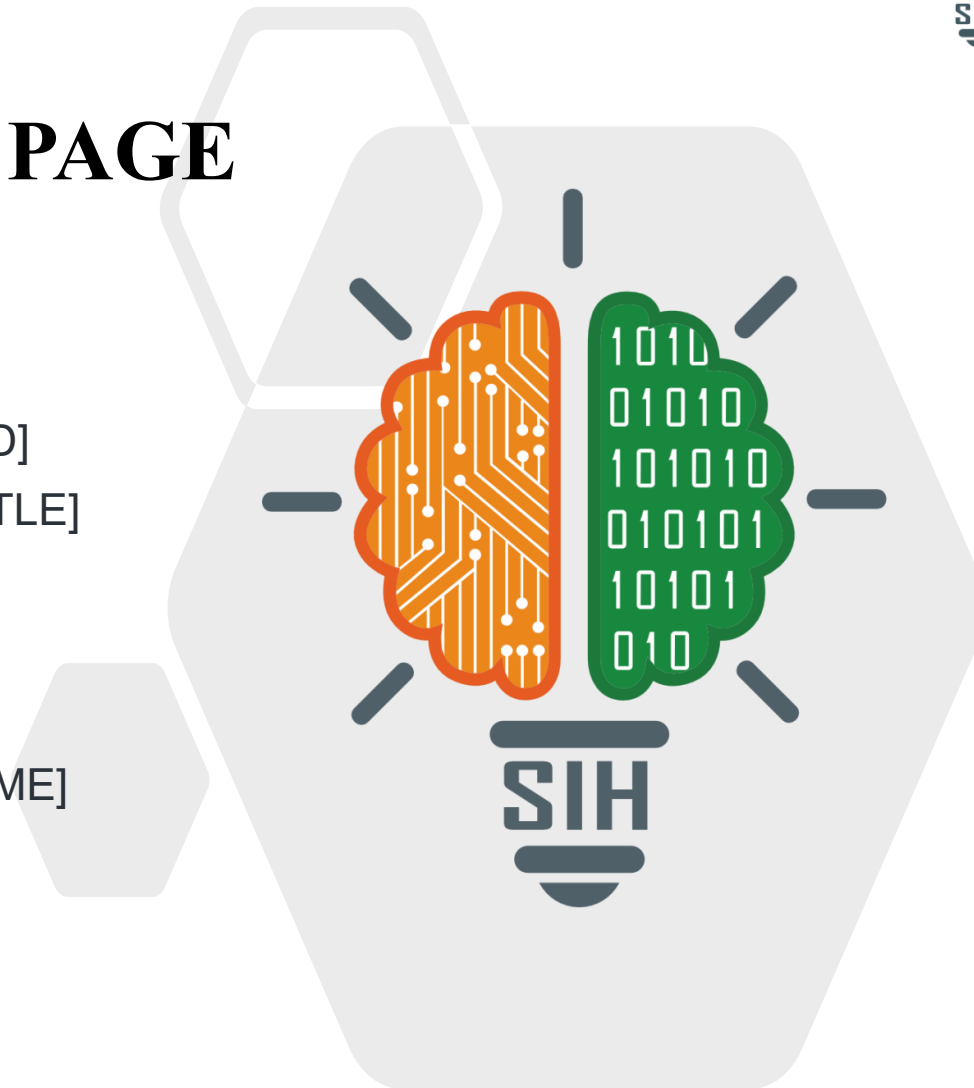
Problem Statement Title: [ENTER EXACT PORTAL TITLE]

Theme: [ENTER PORTAL THEME]

PS Category: Software

Team ID: [ENTER PORTAL TEAM ID]

Registered Team Name: [ENTER EXACT PORTAL NAME]



A well-to-surface decision-support twin for heavy-oil CSS and sucker-rod pump operations.

PROPOSED SOLUTION (DESCRIBE YOUR IDEA / SOLUTION / PROTOTYPE)

A digital twin links CSS reservoir cooling to heavy-oil flow and SRP behaviour in one operator-facing view.

DETAILED EXPLANATION

Thermal state → temperature-dependent viscosity → wellbore resistance → rod/load/vibration indicators → weighted rod-float risk, evidence chain and recommended SPM.
Scenario and what-if screens let staff compare options before acting.

HOW IT ADDRESSES THE PROBLEM

Connects thermal decline and lift mechanics that are often reviewed separately.
Gives the Well Incharge a well-level risk view and the Supervisor a fleet-level context; recommendations remain human-reviewed.

INNOVATION AND UNIQUENESS OF THE SOLUTION

Explainable physics chain + interactive 3D well view + scenario simulation + role-based workflows.
Bundled UI language packs: English, Hindi, Assamese, Telugu and Rajasthani. Bhashini API integration is future work, not live in this build.

Prototype boundary: simulated telemetry; no live OIL/SCADA connection or physical VFD commands.

TECHNICAL APPROACH

TECHNOLOGIES TO BE USED (LANGUAGES / FRAMEWORKS)

React + TypeScript + Three.js | FastAPI + Python/NumPy | REST + WebSocket | Recharts + i18next | Docker Compose

METHODOLOGY AND PROCESS FOR IMPLEMENTATION

Demo or telemetry input → CSS thermal state → temperature-sensitive viscosity → wellbore flow → Gibbs-type damped rod dynamics → weighted fault-risk score → operator recommendation + what-if.



PROTOTYPE TODAY

Physics demo models, scenario controls, REST/WebSocket feeds, risk insights and what-if screens.

NEXT FOR FIELD READINESS

Historian data → calibration → validation → shadow-mode pilot → operator-approved integration.

Numerical/demo physics and weighted risk logic are implemented; trained PINN, Bhashini NMT and field control are not.

FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea • Potential challenges and risks • Strategies for overcoming these challenges

FEASIBILITY

Built with deployable web technologies and modular API/model layers.

Demo exercises thermal, viscosity and rod-dynamics models with simulated scenarios.

Field rollout depends on OIL telemetry access, well configuration and calibration data.

CHALLENGES / RISKS

Simulation-to-field gap; incomplete or noisy sensors.

False alarms or missed events if thresholds are uncalibrated.

OT security, network outages and integration constraints.

Translated technical alerts may be misunderstood.

MITIGATION / PILOT GATES

Replay historian data; calibrate and validate against labeled events.

Start in read-only shadow mode; measure precision, recall and lead time.

Keep human approval; add audit, role controls and stale-data safeguards.

Review language with operators; check Bhashini access/security before integration.

No operational use until field validation, cybersecurity review and operator acceptance are complete.

IMPACT AND BENEFITS

Potential impact on the target audience (field operators, Well Incharges, production engineers, Supervisors) • Social, economic and environmental benefits

OPERATIONAL BENEFIT

Shared well state and risk signals across field and supervisory roles.

Traceable evidence helps explain why a condition is flagged.

What-if comparison supports informed, human-reviewed decisions.

ECONOMIC POTENTIAL

Earlier intervention may prevent avoidable failures and downtime.

Better-informed lift / CSS decisions may protect production and reduce energy waste.

Benefits must be measured in a field pilot; no savings claimed yet.

SOCIAL + ENVIRONMENTAL

Localized interface options can improve accessibility for frontline teams.

If validated, efficient steam and pumping use could lower energy intensity and emissions.

No live Bhashini alert translation or measured emissions reduction in this prototype.

PILOT SUCCESS METRICS

Warning lead time + precision/recall | failures/workovers | uptime + oil rate | kWh per barrel | steam-oil ratio | operator acknowledgement / comprehension

Impacts are hypotheses to test against baseline data, not measured prototype outcomes.

RESEARCH AND REFERENCES



TEAM-SUPPLIED BACKGROUND RESEARCH

"Digital Twin Oil Well Optimization ... Baghewala Field," Gemini Deep Research PDF (18 pp., supplied 28 Sep 2026). Secondary AI-generated synthesis; field figures and claims need OIL-source verification.

[Open supplied research share](#)

CSS THERMAL MODEL FOUNDATION

Boberg, T.C. & Lantz, R.B. (1966), "Calculation of the Production Rate of a Thermally Stimulated Well," SPE Journal. Referenced by the supplied research; confirm final bibliographic metadata before submission.

[Search Crossref title](#)

SRP ROD-DYNAMICS FOUNDATION

Gibbs, S.G. (1963), "Computing Downhole Pump Cards from Surface Dynamometer Cards," SPE Journal 3(1), as cited in rod_dynamics.py. Verify exact citation details before final submission.

[Search title](#)

IMPLEMENTATION / REPRODUCIBILITY

Project source: backend/app/physics (thermal, viscosity, wellbore, rod dynamics and risk detector), backend/main.py (REST/WebSocket demo API), frontend/src (React UI).

[View project repository](#)

Research note: the supplied PDF has no formal bibliography and labels itself AI-generated; validate field statistics and assumptions with Oil India and primary literature.